

Incomplete Markets

Sneha Thankkam Raju, Suryashis Ghosh

23 May 2025

Introduction

This document analyzes an incomplete market economy with unit mass of agents. We follow an Aiyagari-style framework, extending it to incorporate unemployment benefits financed by a proportional wage tax and derive key elements of the agent's problem. Given that the government has a balanced budget every period. The price of capital, r , and the price of labor, w , are determined in a competitive market.

1 Markov Transition Matrix

The employment status of an agent is modeled as a two-state Markov process, where:

- $s = 0$ represents the agent being unemployed.
- $s = 1$ represents the agent being employed.

We are given:

- An unemployed agent becomes employed with probability 0.8.
- An employed agent becomes unemployed with probability 0.3.

The transition matrix P is defined as:

$$P = \begin{bmatrix} \mathbb{P}(s_{t+1} = 0 \mid s_t = 0) & \mathbb{P}(s_{t+1} = 1 \mid s_t = 0) \\ \mathbb{P}(s_{t+1} = 0 \mid s_t = 1) & \mathbb{P}(s_{t+1} = 1 \mid s_t = 1) \end{bmatrix} = \begin{bmatrix} 0.2 & 0.8 \\ 0.3 & 0.7 \end{bmatrix}$$

In tabular format:

	Unemployed $t + 1$	Employed $t + 1$
Unemployed t	0.2	0.8
Employed t	0.3	0.7

2 Expression for Unemployment Benefit b

We are asked to express the unemployment benefit b as a function of the wage rate w , for a given proportional tax rate t , using the law of large numbers and the condition that the government runs a balanced budget.

Step 1: Government Budget Constraint

The government collects tax revenue by imposing a proportional tax t on the wage income of employed agents. It pays unemployment benefits b to agents who are unemployed. Since the government maintains a balanced budget in every period, we have:

$$\text{Total tax revenue on wage income} = \text{Total unemployment benefits}$$

Let N be the mass (or fraction) of employed agents in the population. Since the total population is normalized to 1 (unit mass), the mass of unemployed agents is $1 - N$. The total tax revenue collected from employed agents is:

$$t \cdot w \cdot N$$

and the total benefit payments are:

$$b \cdot (1 - N)$$

Hence, the balanced budget condition is:

$$t \cdot w \cdot N = b \cdot (1 - N) \tag{1}$$

Step 2: Stationary Distribution from the Markov Process

We are given the following Markov transition matrix for employment status:

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.3 & 0.7 \end{bmatrix}$$

Let π_U and π_E denote the long-run (stationary) probabilities of being unemployed and employed, respectively. These satisfy the following conditions:

$$\pi_U + \pi_E = 1 \tag{2}$$

$$\pi_U = 0.2\pi_U + 0.3\pi_E \tag{3}$$

Substituting (2) into (3) to eliminate π_E :

$$\pi_U = 0.2\pi_U + 0.3(1 - \pi_U)$$

$$\pi_U = 0.2\pi_U + 0.3 - 0.3\pi_U = 0.3 - 0.1\pi_U$$

$$1.1\pi_U = 0.3 \quad \Rightarrow \quad \pi_U = \frac{3}{11}$$

Using (2) again:

$$\pi_E = 1 - \pi_U = 1 - \frac{3}{11} = \frac{8}{11}$$

Thus, in the long run:

$$N = \pi_E = \frac{8}{11}, \quad 1 - N = \frac{3}{11}$$

Step 3: Solving for b

Substitute $N = \frac{8}{11}$ into the budget constraint (1):

$$t \cdot w \cdot \frac{8}{11} = b \cdot \frac{3}{11}$$

Multiply both sides by 11:

$$8tw = 3b \quad \Rightarrow \quad b = \frac{8}{3}tw$$

Final Expression

$$b = \frac{8}{3} \cdot t \cdot w$$

This is the expression for the unemployment benefit b as a function of the wage rate w for a given proportional tax rate t .

3 Agent's Value Function $v(k, s)$

The agent maximizes expected lifetime utility over consumption, subject to a budget constraint and the law of capital motion.

Objective Function

The agent maximizes expected discounted utility:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(c_t + \theta)$$

subject to the budget constraint:

$$c_t + i_t = rk_t + ws_t + b(1 - s_t) \tag{1}$$

and the law of motion for capital accumulation:

$$k_{t+1} = (1 - \delta)k_t + i_t \tag{2}$$

Step 1: Solve Equation (2) for i_t :

$$i_t = k_{t+1} - (1 - \delta)k_t \quad (3)$$

Step 2: Substitute Equation (3) into Equation (1):

$$c_t + [k_{t+1} - (1 - \delta)k_t] = rk_t + ws_t + b(1 - s_t)$$

Step 3: Rearranging terms:

$$c_t + k_{t+1} = rk_t + ws_t + b(1 - s_t) + (1 - \delta)k_t$$

Final Form:

$$\boxed{c_t + k_{t+1} = rk_t + ws_t + b(1 - s_t) + (1 - \delta)k_t}$$

where $s_t \in \{0, 1\}$ is the employment status at time t .

Derivation of the Value function

Given, the agent's lifetime utility maximization problem:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(c_t + \theta)$$

subject to the budget constraint (as shown earlier):

$$c_t + k_{t+1} = rk_t + ws_t + b(1 - s_t) + (1 - \delta)k_t$$

From Question 2, we know that the unemployment benefit is:

$$b = \frac{8}{3}tw$$

Step 1: Substitute the expression for b into the budget constraint:

$$c_t + k_{t+1} = rk_t + ws_t + \left(\frac{8t}{3}w\right)(1 - s_t) + (1 - \delta)k_t$$

Step 2: Rearranged to express consumption:

$$c_t = rk_t + ws_t + \left(\frac{8t}{3}w\right)(1 - s_t) + (1 - \delta)k_t - k_{t+1}$$

Step 3: Recursive Formulation

Let the state variables be:

- k : current capital
- $s \in \{0, 1\}$: employment status (1 = employed, 0 = unemployed)

Let $v(k, s)$ be the value function. The agent chooses next-period capital k' (which is k_{t+1}) to maximize current utility plus expected discounted future value:

$$v(k, s) = \max_{k'} \left\{ \ln \left[rk + ws + \left(\frac{8t}{3} w \right) (1 - s) - k' + (1 - \delta)k + \theta \right] + \beta \sum_{s' \in \{0,1\}} P_{ss'} \cdot v(k', s') \right\}$$

4 Capital Distribution Across Agents for Different Tax Rates

The following plot shows the distribution of capital across agents for three different tax rates: $t = 10\%$, 20% , and 30% . As the tax rate increases, agents save less due to lower after-tax wages and higher unemployment benefits. This results in more agents holding lower levels of capital, shifting the distribution toward the lower end.

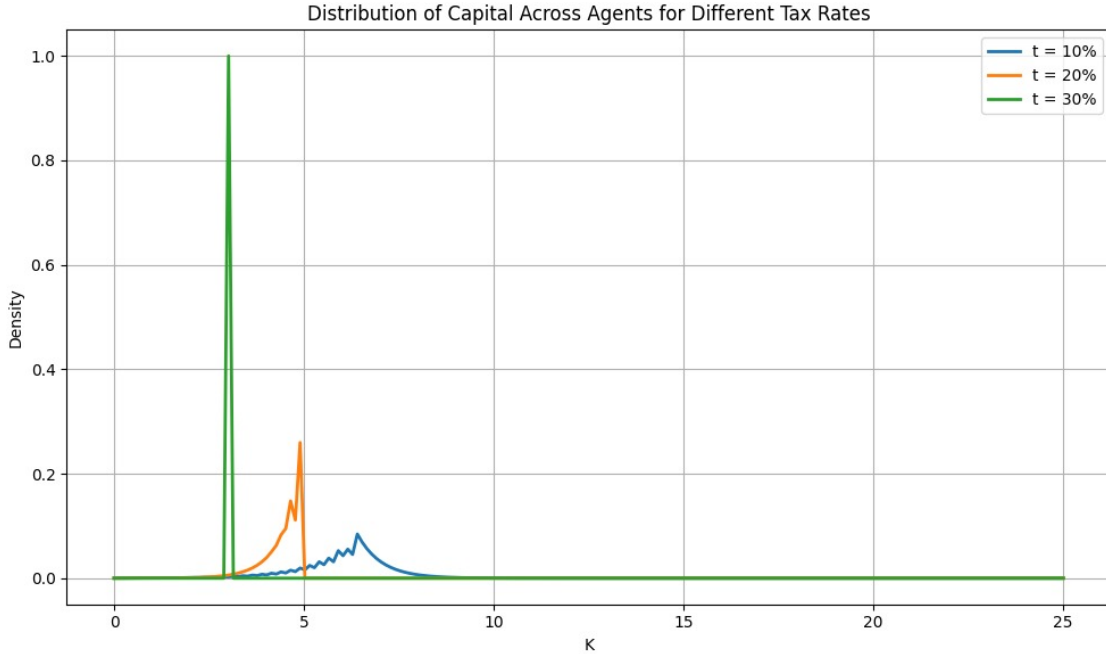


Figure 1: Distribution of Capital Across Agents for $t = 10\%$, 20% , 30%

5 Aggregate Capital Levels for different tax rates

The table below summarizes the aggregate capital K for each of the three tax rates considered. These values are obtained numerically from the model simulation and reflect the average capital holdings across agents in the long-run equilibrium.

Tax Rate (%)	Aggregate Capital K
10%	6.06
20%	4.44
30%	3.02

This reflects the reduced incentive to save when redistribution via unemployment insurance becomes more generous.